Four megatrends, 2026-2045: Tetlock ensemble forecast

Forecast origin: 2026-11-07 | **Panel:** 4 independent efforts x 6 single-skill seats = 24 | **Method:** `.claude/skills/tetlock-ensemble-forecasting`

Bottom line up front

The binding constraint of 2026-2045 is US fiscal arithmetic, not electrification, AI or demography taken one at a time. US net interest reaching 4.5% of GDP in FY2035 is slightly more likely than not: **53% raw mean, 57% median**. All four effort calibrators sit at 54-59%. Fiscal dominance is the modal 2045 end-state at **36%**, narrowly ahead of a "fragmented middle" at **34%**. The US 10-year TIPS yield averages **2.22% in 2035 (P50; P10-P90 0.90-3.40)**: below today's ≈2.9% spike, but well above the 2010s, so secular stagnation does not return by default (**17%**). Trade openness in 2035 is a near coin-flip (**46%** that it is lower than in 2024). Digital services roughly double to **$10.9T**, and China re-reaching 75% of US nominal GDP is a long shot (**19%**).

Scorecard (headline = unextremized raw mean and median)

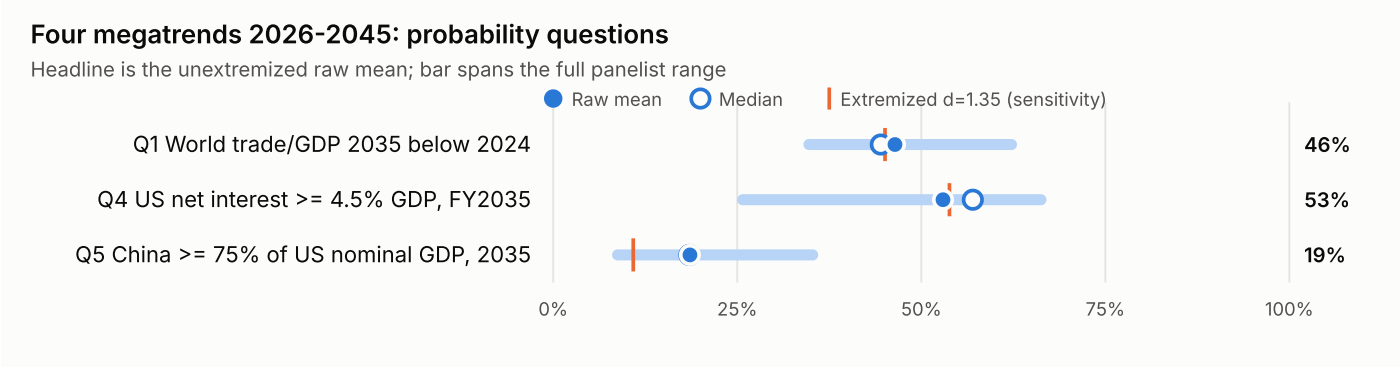
Q	Question (resolves)	Raw mean	Median	Effort range (4 calibrators)	*Sensitivity, d=1.35*
Q1	World trade/GDP 2035 < 2024 (WDI, 2037)	**46%**	45%	41-54%	*45%*
Q4	US net interest >= 4.5% GDP, FY2035 (CBO, 2036)	**53%**	57%	54-59%	*54%*
Q5	China >= 75% of US nominal GDP, 2035 (IMF WEO Oct-2036)	**19%**	19%	17-24%	*11%*
Q6	2045 regime: A fiscal dominance / B capex-led abundance / C secular stagnation / D fragmented middle	**36 / 13 / 17 / 34**	38 / 13 / 16 / 33	A 38-40, B 10-14, C 14-19, D 31-34	-

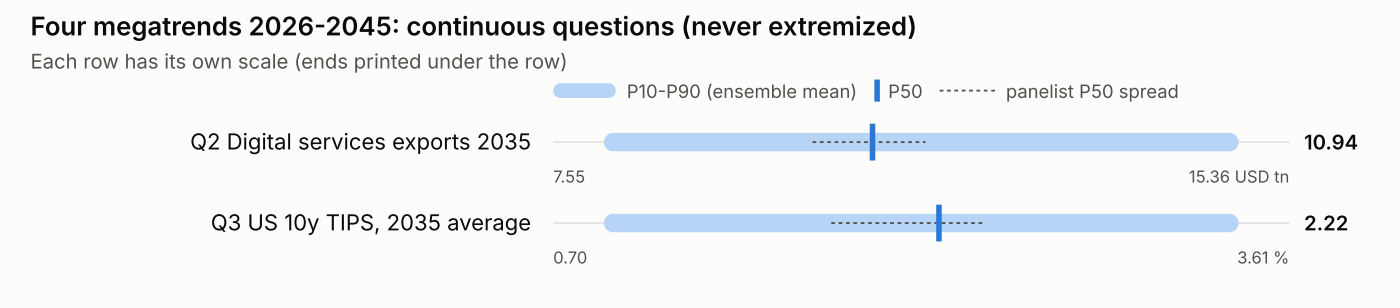
Q	Continuous quantity	Ensemble mean P10 / P50 / P90	Ensemble median P10 / P50 / P90
Q2	Digitally delivered services exports, 2035, USD tn (WTO)	**8.09 / 10.94 / 14.82**	8.10 / 10.90 / 14.75
Q3	US 10y TIPS (`DFII10`), 2035 average, %	**0.90 / 2.22 / 3.40**	0.90 / 2.27 / 3.40

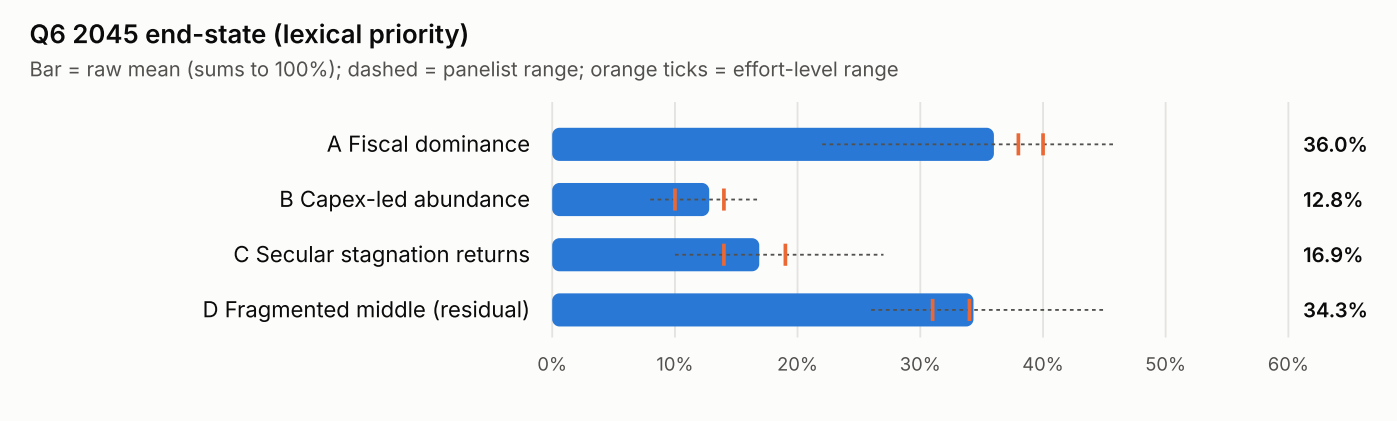
Two signposts that would flip the call.

1. **Where the 10-year settles in 2027-28.** The fiscal lean rests on market yields sitting ≈60-100bp above CBO's assumed 4.1-4.4% path while most debt reprices before 2035. A reversal like October 2023's (≈100bp in ten weeks) that holds would pull Q4 back below 50 and lift C.
2. **The 2027-29 AI-capex refinancing wall.** A bust (red teams put it at 20-35%) lowers rates *and* trade value and lifts the renminbi. A sustained boom does the opposite. It is the single shock that moves Q1, Q3, Q5 and Q6 together.

Biggest measurement caveat. Q1, Q2 and Q5 are nominal-dollar ratios, so exchange rates and commodity prices move them as much as volumes do. Every Wave-1 seat in Efforts 1 and 2 assumed a neutral dollar, and both red teams landed that hit. Also, the 4.5% threshold in Q4 is CBO's own 2035 baseline, so Q4 is a bet on CBO's error, not on the trend.







1. What the panel says about how the four trends interact

Rates and capital scarcity (Q3, Q6-C). All four efforts converge on a real yield that falls back from the 2026 spike but does not return to the 2010s (effort P50s 2.15-2.33%). Three front-loaded demands for capital coincide over 2026-2035:

- AI and datacenter capex, about 1.8% of US GDP in 2026 and rising;
- clean-energy capex, about $2.2T of $3.4T world energy investment;
- deficits widened by lost tariff revenue.

Demography is the swing factor, and the effort split here is the most instructive result. Flow saving falls as populations age (Goodhart-Pradhan), but the *stock* of wealth older households want to hold keeps rising (Auclert, Malmberg, Rognlie and Straub put it at about 200% of GDP by 2100). The demography red team showed that the Wave-1 seats had used only the flow view. That is why C keeps 14-19% rather than collapsing.

Sovereign debt (Q4, Q6-A). The fiscal question is where the trends compound rather than cancel:

- Higher real rates come from capex demand and the term premium.
- The tax base thins: IEEPA tariff receipts are running ≈60% below CBO's February projection. The model cannot show AI eroding the payroll base, because labour share is fixed.
- Ageing raises outlays: OASI is exhausted in Q4 2032 and HI in Q2 2033.

Every effort reversed its own outside-view base rate on Q4, from ≈29% to ≈57%. The cause was the same each time: the record that CBO "almost always overestimates" net interest comes from the 1984-2021 era of falling rates, and that era has ended.

Two red teams added a **regime-label trap**. True fiscal dominance tends to show up as financial repression, with yields capped and debt inflated away. Such a world fails A's 6%-of-GDP test and resolves as C. A is therefore "high measured interest", not "fiscal dominance in general".

Trade (Q1, Q2). Electrification and AI pull the nominal trade ratio in opposite directions:

- Fuel trade shrinks (about -1.3pp of trade/GDP) and goods trade drifts down under fragmentation (about -2 to -4pp).
- Digital services roughly double (about +3 to +5pp).
- AI hardware is now ≈47% of goods-trade *growth* by value, and the WTO's own caveat is that its "high import content" makes trade fragile to an AI slowdown.

Hence a coin-flip on Q1 with a narrow effort range (41-54%). The 2026 Hormuz price spike is transient, and every effort discounted it.

National divergence (Q5). The China/US nominal ratio is governed by the exchange rate more than by real growth:

- China's real-growth lead (about 1.3-1.4pp a year, which the model's PPP leg also shows) is roughly cancelled by its deflator gap.
- The renminbi (about 6.67, its strongest since 2022) is the only route to 0.75, and the PBOC is resisting appreciation beyond about 6.65.
- The outside-view's "no peaked challenger regains its peak" class was shown to be partly selection-biased; Japan set a new high in 1995. That lifted Q5 from ≈11% to ≈19%.

2. Scope and external-benchmark reconciliation

Benchmark	Its scope	This panel's scope	Implication
CBO Feb-2026: net interest 4.6% of GDP in FY2036; 10y assumed 4.1-4.4%	Current law, Feb-2026 rates and tariffs	FY2035 actuals	The threshold sits at CBO's 2035 path, so Q4 prices the sign of CBO's error

Benchmark	Its scope	This panel's scope	Implication
CBO long-term outlook: ≈5.7% in FY2046	Same assumptions, extended	FY2045 >= 6.0% for regime A	The A leg needs CBO to under-project by ≈0.3-0.4pp
WTO "digitally delivered" $5.26T (2025) vs UNCTAD "digitally deliverable" (+7.1%/yr)	Different series	WTO	Not one series. Seats that compounded the UNCTAD rate were down-weighted
NY Fed HLW r* 1.65% (2026Q2)	Model-based short real rate	`DFII10` 10y TIPS	TIPS includes a liquidity and term premium, so a 2035 P50 of 2.2% is consistent with r* ≈1.4-1.6
WDI trade/GDP 56.76% (2024)	Nominal, goods and services	Same; first vintage with 2035	The 2024 base can be revised within the resolving vintage

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3. Wave audit: how the estimates moved

Mean by seat across the four efforts. Waves 2-3 changed the Wave-1 base rates *before* any extremization.

Seat	Q1	Q4	Q5	Q2 P50	Q3 P50	A	B	C	D
A1 outside-view	43	29	12	10.68	1.98	23	14	25	39
A2 fermi (sim-enabled)	54	62	21	10.65	2.32	41	16	13	30
A3 signals	48	62	22	11.15	2.35	35	10	15	41
A4 bayes	42	50	12	11.07	2.30	40	12	14	34
A5 premortem	44	58	26	11.12	2.17	39	13	19	30
A6 calibrator	**48**	**58**	**20**	**10.95**	**2.22**	**39**	**13**	**16**	**33**

Movement from the outside-view base rate to the calibrators, per question:

- **Q4: reversed** (29 to 58). The largest update in every effort, for the same cause in each: the outside-view's reference class is regime-bound.
- **Q6-A: amplified away from base rate** (23 to 39).
- **Q6-C: moderated downward** (25 to 16), with a floor kept by the stock-of-wealth and repression arguments.
- **Q5: moderated toward 50** (12 to 20), after the exchange-rate hit landed.
- **Q1: essentially held** (43 to 48).
- **Q2 and Q3: held**, with bands widened at the top.

The Bayesian seats were the most conservative because they flagged almost every signal as below 2:1. The one signal all four rated above 2:1 was rates sitting above CBO's path.

Simulation discipline held. Every Fermi seat ran the model, adopted only its identified or directional readings, and refused its levels:

- Its 2040-45 world growth (3.58% at baseline) runs ≈0.5pp above IMF's medium-term path, so no seat put B above 17%.
- Its "fragmentation raises interest" reading was shown to be circular: it comes only from the lever's own defence deficit.
- Its "AI adds +0.04 to interest/GDP" reading follows from the model's construction, not from evidence.

No calibrator's number equals the model's number.

On extremization. Q5 shows why `d = 1.35` is only a sensitivity column. Pooling in log-odds drags a low-probability question further down (19% to 11%), and these 24 seats share model priors and overlapping sources, which violates the independence that extremization assumes. Fit `d` on this ensemble's record once questions resolve (skill Step 6).

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4. Effort-by-effort synthesis (calibrator view)

Effort	Q1	Q4	Q5	Q2 P50	Q3 P50	A / B / C / D	Largest update (cause)
1 Energy and electrification	53	58	18	11.2	2.20	39 / 13 / 17 / 31	Q4 +33: threshold equals CBO baseline; yields 60-90bp above CBO path; tariff ruling adds ≈$2T to deficits
2 AI and robotics	43	54	17	11.1	2.33	38 / 14 / 14 / 34	Q4 +24: most debt reprices before FY2035 at above-path yields
3 Demography and OLG	54	59	19	10.6	2.15	40 / 10 / 19 / 31	Q4 +29: verified 60% tariff-revenue shortfall and above-path real yields
4 Fragmentation and fiscal	41	59	24	10.9	2.20	38 / 13 / 15 / 34	Q4 +29; model's fragmentation-to-interest reading given zero weight (circular)

Effort-specific contributions that survived synthesis:

- **Effort 1:** fuels are ≈6pp of the trade ratio and will lose value. The model's "a high fossil price raises growth" is a closure artefact, because it has no terms-of-trade loss.
- **Effort 2:** the AI-capex bust tail is fat (25-35%), and AI goods make trade fragile. The "both AI tails hurt the budget" argument was judged posed, not shown.
- **Effort 3:** the flow-versus-stock ageing debate keeps C alive, and CBO's demographic outlook already assumes deaths exceed births from 2030.
- **Effort 4:** the political reaction function (budget deals, repression, a post-Warsh Fed) is the main reason A is not higher. The October 2023 reversal is the anti-anchoring analogue.

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5. Observable Bayesian signposts

Signpost (check annually)	Moves	Direction
10y Treasury vs CBO path; ACM term premium (0.98% in Oct-2026)	Q4, Q3, A, C	Persistent above-path, toward Q4 YES and A; reversal toward C
Tariff legality and receipts vs CBO (2026: -60%)	Q4, A	Replacement revenue lowers Q4
Hyperscaler capex and free cash flow; AI credit spreads (CoreWeave 2031s ≈79c)	Q1, Q2, Q3, Q5, B	Bust: Q3 and Q1-value down, Q5 up
USD/CNY and China's GDP deflator (turned positive in 2026Q2 per secondary estimates)	Q5	Renminbi below ≈6.3 sustained: Q5 above 35%
WTO digitally delivered services growth vs ≈7.5%/yr	Q2, Q1	
SS/HI trust-fund legislation before 2032	Q4, A	General-revenue transfer raises A
G7 long-end stress (JGB 30y ≈4.2%, gilt 10y 5.5%, French CDS)	A (G7 leg)	

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6. Provenance, limits and next steps

- **Records:**
  - `questions.md` (locked before forecasting, commit `3bee958`);
  - `adapter-spec.json` and `adapter-output.*` (the published model-to-question adapter);
  - `seats/` (all 24 replies verbatim, and the Fermi seats' extra specs in `seats/fermi-specs/`; the seats cite their original scratch paths);
  - `bundle.json`, rebuilt from `seats/` by `build_bundle.py`;
  - `aggregate/`.
- **Model version:** the adapter readings in `adapter-output.*` were generated at model commit `c06e55d`, before PR #95 (demand-led new build, idle-fossil fix, China catch-up calibration, scenario carbon prices) was merged. Re-running the same spec on the merged model leaves the baseline readings almost unchanged (regime B's 2040-45 growth leg 3.58% → 3.60%). The named stress scenarios grow faster: tech-stagnation, debt-populism and net-zero gain up to +0.7pp/yr of 2026-35 growth for both the US and China, and debt-populism's 2045 interest share falls 12.2% → 11.1%. Seats used those scenarios only as directional context, so no panel number depends on the change.
- **Orchestration:** the skill prefers the Workflow tool. This container has 4 CPUs, which caps a workflow at 2 concurrent agents, so the waves ran as parallel background agents instead (12, then 8, then 4). Efforts pipelined independently, and each Wave-2/3 seat read only its own effort's packet.
- **Source quality:** several seats could not reach `wto.org`, `crfb.org` or `api.worldbank.org`, so some figures are secondary. These are flagged in each seat's caveats: the HLW r*, the China deflator turning positive, and the Japan 1988-95 ratios.
- **Auditability, not proven accuracy:** role separation makes each move traceable. It is not evidence of better Brier scores; that needs resolved outcomes.
- **Not yet done (skill Step 6):** sealing and registering these questions in `packages/forecast-workbench`, so they are scored on resolution and `d` can be fitted.

Sources (orchestrator baselines)

- CBO, *Budget and Economic Outlook 2026-2036*: https://www.cbo.gov/publication/61882 (via CRFB: https://www.crfb.org/papers/cbos-february-2026-budget-and-economic-outlook)
- CBO Monthly Budget Review, Sep-2026: https://www.cbo.gov/system/files/2026-10/61985-MBR.pdf
- FRED `DFII10` (via https://www.thetrading.tools/real-yields); TIPS reopening: https://tipswatch.com/2026/09/17/10-year-tips-reopening-gets-real-yield-of-2-653-highest-in-nearly-18-years/
- World Bank WDI trade/GDP (via https://ycharts.com/indicators/world_trade, https://statista.com/chart/29597/total-trade-volume-as-a-percentage-of-gross-domestic-product)
- WTO Global Trade Outlook, 8 Oct 2026: https://www.wto.org/english/news_e/news26_e/stat_08oct26_510_e.htm
- WTO digitally delivered services 2025 (via JETRO): https://www.jetro.go.jp/world/gtir/2026/ch1/sec2/sub1/
- IMF WEO Oct-2026: https://www.imf.org/-/media/files/publications/weo/2026/october/english/ch1.pdf

- NY Fed HLW r* (secondary): https://www.ibtimes.sg/A000OGx

Each seat's own sources are inline in `seats/`.